# Backdoor Defense Strategy for Image Classification Tasks Based on Frequency Domain Perturbation

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***Abstract*—Deep neural networks (DNNs) have achieved significant progress but face growing security threats, particularly backdoor attacks. Adversaries implant backdoors into pretrained models and use specific triggers to activate the backdoor, inducing the model to output incorrect results. To cope with backdoor attacks, we systematically introduce frequency domain perturbations into backdoor defense scenarios. By preserving phase semantics and perturbing amplitude high-frequency features, the repaired model will shift from trigger dependency to semantic dependency, thereby weakening the coupling relationship between backdoor trigger patterns and target categories. We propose Freq-Pret to demonstrate this backdoor defense strategy, which is a novel backdoor defense scheme combining frequency-domain perturbations with lightweight retraining. Theoretical analysis and experiments demonstrate that Freq-Pret effectively repairs backdoors without compromising initial accuracy, enabling the repaired model to resist attacks and correctly categorize contaminated samples with high probability. Compared to existing methods, Freq-Pret offers clear advantages.**

***Index Terms*—Backdoor attack defense, deep neural networks (DNNs), pretrained model security.**

## I. Introduction

DEEP neural networks (DNNs) have been widely applied in many different fields, such as face recognition and autonomous vehicles. However, the accompanying security issues have also attracted widespread attention [1], with backdoor attacks being particularly prominent [2], [3]. In DNN backdoor attack [4], the adversary injects a backdoor into the victim model, which can be achieved by poisoning the training datasets [5], [6] or modifying neuron parameters [7], [8]. These backdoor models behave normally when receiving benign samples, but will erroneously output specific labels desired by the adversary when receiving an attacker-specified trigger [9], [10]. Backdoor attacks are often very covert. For example, in facial recognition systems with backdoors [11], [12], attackers can simply activate attacks by wearing/holding physical trigger objects (such as sunglasses, white tape, bandana, and earrings).

Adversaries have many opportunities to launch backdoor attacks. Most ordinary users do not have the computing resources and training data required for DNN model training and have to rely on third-party platforms and/or use third-party datasets from unknown sources for model training, or even directly deploy pretrained models provided by third parties. Malicious third parties can implant backdoors by contaminating datasets, manipulating the training process, and even directly modifying model parameters.

Various strategies have been proposed to defense against backdoor attacks, including sample preprocessing [13], [14], [15], model diagnosis [16], [17], [18], [19], [20], [21], [22], model reconstruction [23], [24], [25], [26], [27], and trigger synthesis [28], [29]. However, they have their own limitations and face various challenges in practical applications. For example, model reconstruction base schemes such as fine pruning (FP) [23] can effectively defense backdoors in theory, but in practice, they often weaken the performance of the model on the original task due to excessive weight reduction during the pruning process; The trigger synthesis strategy represented by DeepInspect [28] attempts to reconstruct triggers through reverse engineering, which requires enormous computational resources during their execution.

To mitigate backdoor attacks, we propose a defense strategy using frequency-domain perturbation. Existing research on frequency-domain backdoor mainly focuses on the perspective of attacks. Zeng et al. [30] proposed a practical way to create smooth backdoor triggers. FIBA [31] implants hidden backdoors triggers by fine-tuning the frequency components. This article systematically introduces frequency-domain perturbations into backdoor defense scenarios. By preserving phase semantics and perturbing amplitude high-frequency features, the repaired model will shift from trigger dependency to semantic dependency, thereby weakening the coupling relationship between backdoor trigger patterns and target categories. We propose Freq-Pret to demonstrate this backdoor defense strategy, which is a novel backdoor defense scheme combining frequency-domain perturbations with lightweight retraining. The main contributions are summarized as follows.

1) We propose a backdoor defense strategy based on frequency-domain perturbation. It guides the model to focus on the high-level semantic information and ignore the backdoor trigger.

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2) We design a lightweight backdoor defense scheme. It combines frequency-domain perturbation with lightweight retraining, which effectively defends against backdoor attacks without sacrificing accuracy.
3) We analyze the proposed scheme from the theoretical perspective and quantitatively evaluate it from the experimental perspective.

The rest is organized as follows. Section II contains related works. Section III provides details of the proposed scheme. Section IV consists of theoretical analyses. Section V presents the experimental evaluation. Section VI is the conclusion.

## II. Related Work

### A. Backdoor Attacks

Early backdoor attacks on artificial neural networks include BadNets [2] and TrojanNN [7], which implant special triggers into training sets to manipulate model outputs. Chen et al. [3] confirmed that just a few contaminated samples can successfully implant backdoors. Barni et al. [5] embedded imperceptible sine or ramp signals in target samples to achieve preset outputs without modifying labels.

With defense advancements, attackers innovate their covert methods. Backdoor triggers evolved from simple explicit small blocks to diverse and covert forms like noise, deformation, frequency injection, and composite features, posing greater defense challenges. WaNet [6] uses image deformation as triggers. Li et al. [10] combined steganography and regularization for invisible attacks. Lin et al. [9] proposed composite attacks using benign features. Feng et al. [31] performed frequency-based attacks on medical images. Li et al. [4] introduced customized invisible attacks. Zeng et al. [30] analyzed high-frequency artifacts in triggers, proposing smooth trigger generation to improve concealment.

### B. Backdoor Defense

Sample preprocessing-based schemes modify input samples to mitigate trigger effects. DeepSweep [13] uses data augmentation and model fine-tuning to alter decision boundaries, weakening backdoor influence. Februus [14] purifies data by removing and restoring contaminated inputs. However, DeepSweep requires multiple experiments to find the best augmentation scheme, while Februus consumes significant computing resources. PnA [32] addresses data poisoning in federated learning, enhancing model security through robust aggregation. Chen et al. [33] leveraged contaminated sample sensitivity for defense. Recent studies focus on dataset purification [34] and reprogramming preprocessing [35].

Model diagnosis-based schemes analyze model output or internal dynamics to detect backdoor influence. Neural Cleanse [16] identifies anomalies by label indices. ABS [17] locates manipulated neurons via activation value spikes. MNTD [18] and RED [19] use meta network analysis and reverse engineering to detect Trojan infections. Wang et al. [20] performed feature space analysis for comprehensive trigger detection and reverse design. Hou et al. [36] proposed input-level backdoor detection via parameter-oriented scaling consistency.

Model reconstruction-based schemes eliminate backdoor traces by adjusting the model architecture or training process. Fine-tuning (FT) [23] reduces neurons that are inactive on pure data, theoretically removing backdoors. MCR [25] uses the pattern connectivity of loss landscapes to study the adversarial nature of neural networks, and mitigates the impact of adversarial samples on the model by optimizing the connections through real data paths. NAD [26] uses teacher models to guide student models in training backdoor models, eliminating backdoors through knowledge distillation. Zheng et al. [27] remove backdoors by evaluating the Lipschitz constants of each channel in the neural network. Xia et al. [37] used attention graph distillation to eliminate backdoor triggers in DNNs. BadCleaner [38] uses multiteacher distillation and attention mechanisms to defend against backdoor attacks in federated learning. Chen et al. [39] used adversarial samples to defend against backdoor attacks in federated learning, demonstrating its effectiveness in various attack scenarios.

Trigger synthesis-based schemes identify and reconstruct backdoor triggers to eliminate backdoor threats. For example, DeepInspect [28] reconstructs the training set through model inversion, aiming to reveal and reconstruct backdoor triggers. Huang et al. [29] improved the robustness and defense capability of the model by extracting and distilling cognitive backdoor patterns from images.

In addition, some researchers have constructed optimized defense frameworks for specific scenarios. MoCC-BD-FID [40] focuses on intrusion detection of federal industrial control systems. DACO-BD [41] focuses on backdoor defense in synthetic aperture radar image classification.

Overall, sample preprocessing-based schemes are simple to implement but may face potential incomplete sample processing in practical applications. Model diagnosis-based schemes can effectively verify whether the model has been attacked, but often come with high computational overhead. Model reconstruction-based schemes can eliminate backdoor threats to some extent, but the model reconstruction may also sacrifice the model's performance. Trigger synthesis-based defense schemes have shown significant effectiveness in backdoor attacks; they also face extremely high demands for computing resources, which undoubtedly set certain obstacles for their widespread application. Scenario-specific defense schemes perform well on specific scenarios, but may not necessarily be generalizable to other scenarios.

### C. Relationship Between the CNN and the Frequency Domain

Chen et al. [42] pointed out that CNNs typically rely on high-frequency components in the amplitude spectrum of images, which makes the model susceptible to noise and other forms of interference. Xu et al. [43] confirmed that by learning in the frequency domain, the CNN can more efficiently analyze image data and reduce sensitivity to high-frequency interference. FIBA [31] implants hidden backdoors by fine-tuning the frequency components of images, highlighting the limitations of traditional defense measures based on image content analysis in identifying frequency-domain tampering. Liu et al.'s [44] research indicates that compared

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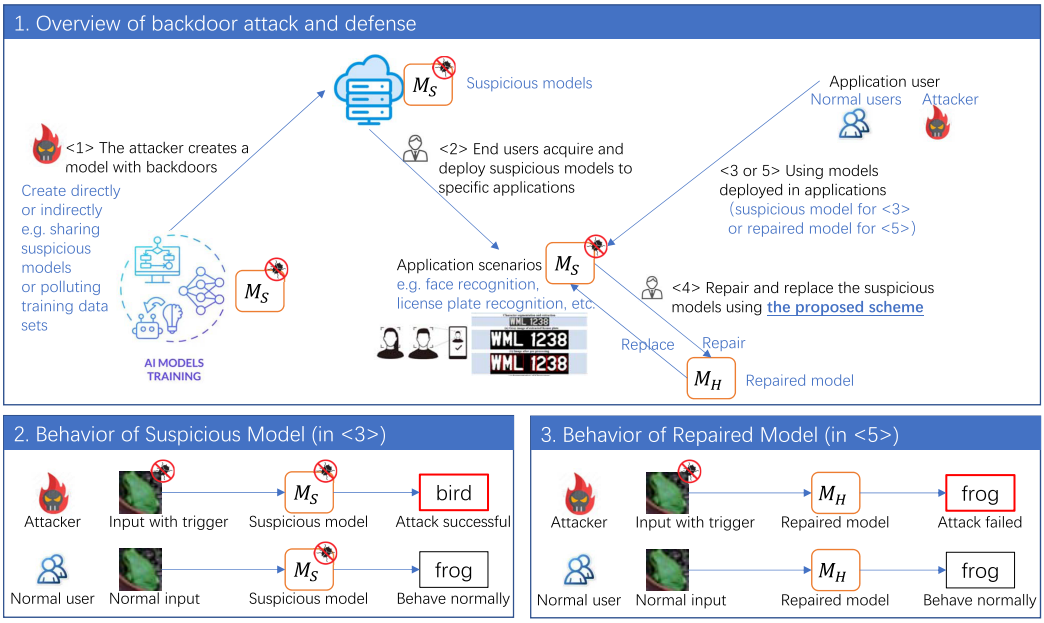



Fig. 1. Application scenario demonstration of the proposed scheme.

to amplitude spectra, phase spectra contain more information about image structure and high-level semantics, which are crucial for identifying the authenticity and forgery of images.

Existing research indicates that frequency-domain analysis has great potential for enhancing the functionality and security of CNNs, and frequency-domain features play a key role in improving CNN performance, thus increasing model interpretability and robustness.

## III. Frequency-Domain Perturbation-Based Backdoor Defense Scheme

This section will provide a detailed introduction to the proposed scheme and analyze its limitations.

### A. Threat Model and Basic Hypotheses

We define the threat model and the capabilities of attackers through two basic hypotheses.

*Hypothesis 1:* It assumes that the attacker can create a model with backdoors and deliver it to the end users through some means.

The model training of DNNs usually requires a large amount of training samples and computing resources, which are difficult for ordinary users to afford. As an ordinary user, it is common to upload data to third-party platforms for model training, or use suspicious datasets provided by third parties for model training, or even download pretrained models that have not been reviewed and verified.

*Hypothesis 2:* It assumes that attackers are unable to intervene when users perform backdoor repairs on suspicious models, as well as when deploying and using repaired models.

Once the user downloads the suspicious model locally, they have control over the local model. As long as the user handles it properly, whether it is during the repair phase, deployment phase, or usage phase, attackers cannot make any modifications to the user's local model again.

*Hypothesis 3:* After deploying the repaired model, attackers can use samples containing backdoor triggers as input to launch backdoor attacks on the deployed model.

### B. Overview

Fig. 1 describes the typical application scenario of the proposed scheme for defending against backdoor attacks of the DNN. Part 1 of Fig. 1 is the overview of backdoor attack and defense, which is divided into five steps. In step 1, the adversary creates a model with backdoors directly or indirectly. For example, the adversary pollutes the dataset for model training, or the adversary invades the model training platform, or the adversary attacks the model hosting platform. In step 2, the end user acquires the suspicious model $M_s$ and deploys it into specific application scenarios, such as a face recognition system, a license plate recognition system, and so on. In step 3, we assume that both adversary and normal users can use the suspicious model deployed in the application scenario. The adversary attempts to activate the backdoor in the suspicious model by using the malicious sample containing the trigger as input, thus completing the backdoor attack. In step 4, the proposed scheme is adopted to implement backdoor defense. The suspicious model $M_s$ will be replaced by the repaired model $M_h$. The difference between step 5 and step 3 is that the adversary and normal users access the repaired model $M_h$ instead of the suspicious model $M_s$. Parts 2 and 3 of Fig. 1 demonstrate the behavior of the suspicious model $M_s$ in step 3 and the behavior of the repaired model $M_h$ in

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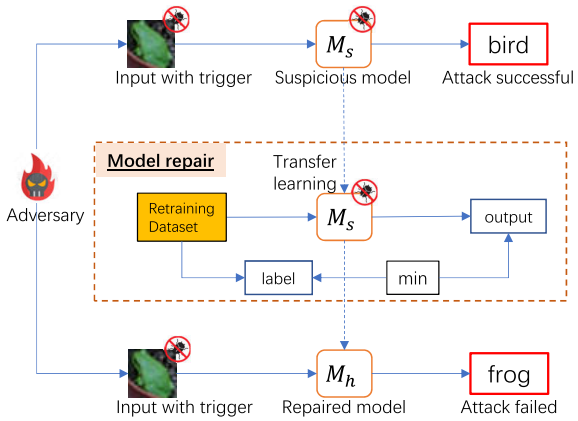



Fig. 2. Overview of the proposed scheme.

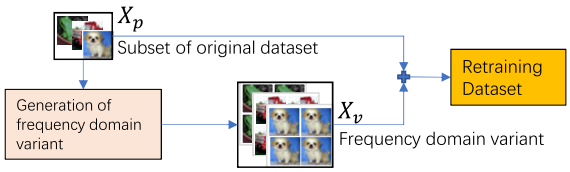



Fig. 3. Retraining dataset.

step 5, respectively. For the sample (frog) containing a trigger, $M_s$ outputs an incorrect result (bird) preset by the adversary. The normal samples do not activate the backdoor of the suspicious model $M_s$ and thus output the correct result (frog), which is also the concealment of backdoor attacks. For the repaired model $M_h$, although the input samples may contain triggers, the triggers did not work, and the backdoor attack fails.

Fig. 2 is the overview of the proposed neural network backdoor attack defense scheme. The potential backdoor in the suspicious model $M_s$ is repaired by a specific lightweight retraining process based on the retraining dataset.

### C. Retraining Dataset

The retraining dataset (Fig. 3) consists of two parts, namely the subset $X_p$ of the original dataset and the frequency-domain variants $X_v$.

*1) Subset $X_p$:* The proposed scheme assumes that a smaller subset $X_p$ of the original dataset is available before model repairing. If the users train suspicious models on a third-party platform with their own dataset, they already have the original dataset. For pretrained models, those trained on publicly available datasets typically indicate the source and download link. Models trained on nonpublic datasets usually provide a sample dataset or training set composition rules when shared.

The sample ratio $\rho$ can be expressed as

$$\arg\min_{\rho}\{f(Cost(\rho), ACC(\rho), RA(\rho), ASR(\rho))\} \tag{1}$$

$f$ is a function that comprehensively considers the training cost and the defense performance. $Cost(\rho)$ is the model retraining cost, which is a monotonically increasing function of $\rho$. ACC (clean accuracy, i.e., the prediction accuracy of clean/normal samples), robust accuracy (RA) (RA, i.e., the prediction accuracy of poisoned samples to the original class), and attack success rate (ASR, i.e., the prediction accuracy of poisoned samples to the target class) are used to evaluate defense performance. $ACC(\rho)$ and $RA(\rho)$ are monotonically increasing functions of $\rho$; and $ASR(\rho)$ is a monotonically decreasing function of $\rho$. The smaller ASR and larger ACC and RA indicate better defense performance.

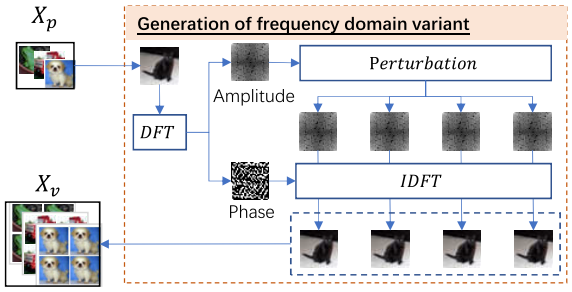



Fig. 4. Generation of the frequency-domain variant.

*2) Subset $X_v$:* The frequency-domain variant dataset $X_v$ is generated based on the subset $X_p$. Fig. 4 describes the generation process of frequency-domain variant samples, which will be detailed in Section III-D.

### D. Generation of Frequency-Domain Variant

As shown in Fig. 4, it mainly includes three core steps: 1) extract amplitude spectrum and phase spectrum (DFT); 2) add perturbation to the high-frequency part of the amplitude spectrum (perturbation); 3) synthesize variant image (IDFT).

*1) Extract Amplitude and Phase Spectrum:* Each sample of the subset $X_p$ can be decomposed into amplitude spectrum and phase spectrum by using the discrete Fourier transform (DFT), which can be implemented through the fast Fourier transform (FFT) in practice. For a given image f$(x, y)$ of size $W \times H$, the frequency-domain image $F(u, v)$ can be obtained by a 2-D DTF, defined as follows:

$$F(u,v) = \sum_{x=0}^{W-1}\sum_{y=0}^{H-1} f(x,y)e^{-i2\pi\left(\frac{ux}{W}+\frac{vy}{H}\right)}. \tag{2}$$

Among them, $(x, y)$ represents the coordinates in the spatial domain, used to describe the spatial position of the image. $(u, v)$ represents the coordinates in the frequency domain, used to describe the frequency components of an image. $e^{i\theta}$ represents the complex exponential term in Euler's formula, where $i = \sqrt{-1}$ and $\theta$ is the angle.

In the context of the DFT, the magnitude spectrum and phase spectrum can be derived from the frequency-domain representation $F(u, v)$.

The magnitude spectrum can be obtained by taking the absolute value of the complex frequency-domain value

$$A(u,v) = |F(u,v)|. \tag{3}$$

The phase spectrum can be obtained by extracting the phase angle of the complex frequency-domain value

$$\phi(u,v) = \arg(F(u,v)). \tag{4}$$

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*2) Add Perturbation:* The perturbation used to construct the frequency-domain variant sample will be added to the high-frequency part of the amplitude spectrum.

First, set the frequency range $[F_{\min}, F_{\max}]$ as the high-frequency region, which is recommended to set one-quarter to one-half of the frequency-domain image size.

Then, construct a frequency mask $M$, which ensures that only amplitude values within a specified frequency range are affected by perturbation. The size of the mask is the same as the frequency-domain image size obtained by DFT. The mask takes a value of 1 in the specified high-frequency region and 0 in other regions. The definition is as follows:

$$M(u,v) = \begin{cases} 1, & F_{\min} \le \sqrt{\left(u-\frac{W}{2}\right)^2 + \left(v-\frac{H}{2}\right)^2} \le F_{\max} \\ 0, & otherwise. \end{cases} \tag{5}$$

Next, construct a random perturbation matrix $R$, for example,

$$R(u,v) \sim U(-1,1) \tag{6}$$

where $U(-1,1)$ is the uniform distribution.

Finally, the adjusted amplitude spectrum $A'(u,v)$ is obtained by disturbing the original amplitude spectrum $A(u,v)$ as follows:

$$A'(u,v) = A(u,v)(1 + \lambda\beta MR(u,v)). \tag{7}$$

The mixing degree $\lambda$ is used to generate multiple frequency-domain variant samples from the same image, for example, setting $\lambda \in \{0.1, 0.2, 0.3, 0.4\}$ to generate 4 variant samples.

The noise ratio $\beta$ controls disturbance intensity, expressed as

$$\arg\min_{\beta}\{h(ACC(\beta), RA(\beta), ASR(\beta))\}. \tag{8}$$

$ACC(\beta)$ is minimally affected by $\beta$. $RA(\beta)$ are monotonically increasing functions of $\beta$; $ASR(\beta)$ is a monotonically decreasing function of $\beta$. $h$ describes the combined effects of them.

*3) Synthesize Variant Image:* Given the phase spectrum $\phi(u,v)$ and the adjusted amplitude spectrum $A'(u,v)$, the inverse discrete Fourier transform (IDFT) can be used to reconstruct the noisy disturbance image, using the following steps.

First, construct the complex spectrum $F(u,v)$ by combining the magnitude spectrum $A'(u,v)$ and phase spectrum $\phi(u,v)$

$$F(u,v) = A'(u,v) \cdot e^{i\phi(u,v)}. \tag{9}$$

Then, apply the IDFT to the complex spectrum $F(u,v)$ to obtain the spatial-domain representation of the image

$$f(x,y) = \mathcal{F}^{-1}[F(u,v)]. \tag{10}$$

Among them, $\mathcal{F}^{-1}$ represents the IDFT. Given a frequency-domain image $F(u,v)$ of size $W \times H$, its 2-D IDFT is defined as

$$f(x,y) = \frac{1}{WH}\sum_{u=0}^{W-1}\sum_{v=0}^{H-1} F(u,v)e^{i2\pi\left(\frac{ux}{W}+\frac{vy}{H}\right)}. \tag{11}$$

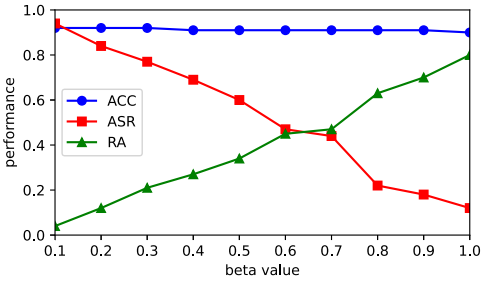



Fig. 5. Impact of the noise ratio on CIFAR-10.

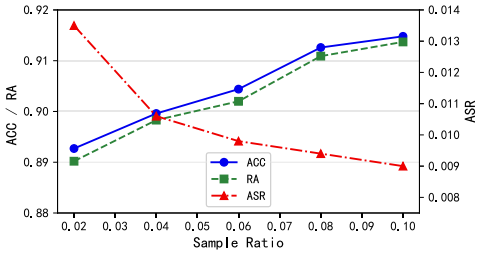



Fig. 6. Impact of the sample ratio.

## E. Repair of Suspicious Models

The repair process of the suspicious model $M_s$ is defined as

$$\mathcal{L} = \sum_{(x_i,y_i)\in X_p \cup X_v} \mathcal{J}(M_h(x_i;\theta), y_i). \tag{12}$$

The training data consists of $X_p$ and $X_v$. $y_i$ is ground truth for $x_i$. The model parameter $\theta$ is initialized with the parameters of $M_s$. According to transfer learning theory, the model repair process will be a lightweight retraining process.

To further expedite the retraining process, we freeze certain layers of the suspicious model $M_s$ before the retraining process begins. The loss function can be defined as

$$\mathcal{L} = \sum_{(x_i,y_i)\in X_p \cup X_v} \mathcal{J}(M_h(x_i;\theta_{train}), y_i). \tag{13}$$

The trainable parameters are $\theta_{train} = \{\theta_j : j \notin \mathcal{F}\}$, where $F$ contains the indices of the layers whose parameters are frozen. We will freeze the initial layer weight parameters of the network. According to deep learning theory, these shallow layers are mainly responsible for capturing the basic components of the image and some simple low-level features, which often have low correlation with specific backdoor triggers.

## F. Practical Deployment Guidance on $\rho$ and $\beta$

*1) Setting of the Sample Ratio $\rho$:* In actual deployment, users usually only have a limited number of samples available for repair. More retraining samples also mean an increase in training costs. We cannot blindly pursue a larger sample size. According to Section V-D, when $\rho$ increases to about 0.06, there is a significant marginal decrease in the improvement of the ACC, RA, and ASR. Therefore, considering both defense effectiveness and retraining costs, it is recommended to prioritize selecting a smaller sample ratio (e.g., 0.03~0.06),

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which can usually achieve relatively stable repair results. If the deployment scenario has stronger constraints on data or computing power, it can start from a smaller scale and gradually increase until the performance reaches an acceptable level.

*2) Setting of the Noise Ratio $\beta$:* The emphasis on accuracy and security varies in different application scenarios. We recommend setting $\beta \in [0.1, 1]$. $\beta = 0.1$ is considered a mild defense configuration that can defend against general backdoor attacks. $\beta = 1$ is considered an enhanced defense configuration to deal with stealthier attacks. For application scenarios that focus on the ACC, set $\beta$ close to 0.1, and for application scenarios that focus on the ASR, set $\beta$ close to 1.

In actual deployment, a progressive strategy can be used to set $\beta$. On the premise that the performance indicators of the repaired model meet the requirements of the application scenario, gradually increase $\beta$ and perform incremental repairs on the suspected model to enhance its ability to mitigate more covert backdoors. First, a mild configuration (such as $\beta = 0.1$) is used to construct an initial retraining dataset for repair, and the ACC changes of the repaired model are verified. Then, gradually increase $\beta$ to expand the retraining dataset, continue to perform incremental repairs on the model, and monitor performance metrics until a suitable $\beta$ is selected based on the task's tolerance for performance metrics.

## IV. Theoretical Analysis

Theoretical analysis is conducted from three aspects: defense principles, lightweight retraining, and limitations.

### A. Analysis of Defense Principles

From the frequency-domain perspective, backdoor triggers are mainly embedded in the high-frequency amplitude spectrum. Zeng et al. [30] found that existing triggers exhibit severe high-frequency artifacts, which persist across different datasets and resolutions. Suspicious model makes backdoor attacks possible by detecting the feature of backdoor triggers in the high-frequency amplitude spectrum.

The proposed scheme guides the repaired model to ignore the high-frequency amplitude spectrum, rendering the backdoor trigger information embedded in these high-frequency amplitude spectra ineffective on the repaired model. We construct frequency-domain variants $X_v$ by introducing noise into the high-frequency amplitude spectrum. The added noise disrupts the high-frequency amplitude spectrum in $X_v$. Meanwhile, the high-frequency amplitude spectrum in $X_p$ remains intact. These two types of samples coexist, and their labels are consistent, aiming to inform the model training process that the high-frequency amplitude spectra are unhelpful for classification and should be ignored. Consequently, the repaired model will ignore the high-frequency amplitude spectrum features of the backdoor trigger, rendering the backdoor trigger ineffective. This design is also consistent with the spectral bias of neural networks [45], where high-frequency components are more sensitive to perturbations.

Ignoring the high-frequency amplitude spectrum will not significantly reduce the classification performance of the repair model for normal samples. CNN classification mainly relies on the phase spectrum, rather than the high-frequency part of the amplitude spectrum. Chen et al. [42] points out that excessive reliance on amplitude spectra can actually make the model susceptible to interference, such as noise and adversarial attacks. Guiding the repaired model to abandon the high-frequency amplitude spectrum may also enhance the robustness of the repaired model.

TABLE I
Performance Comparison of Different Defense Schemes (Group 1)

| Dataset | Backdoor Model | Ours ($\beta$=0.1) | | | Ours ($\beta$=1) | | |
|---|---|---|---|---|---|---|---|
| | | ACC | ASR | RA | ACC | ASR | RA |
| CIFAR-100 | SIG | 0.69 | 0.66 | 0.11 | 0.60 | 0.25 | 0.13 |
| | ISSBA | 0.67 | 0.64 | 0.17 | 0.62 | 0.16 | 0.32 |
| | Blended | 0.67 | 0.93 | 0.04 | 0.57 | 0.00 | 0.57 |
| | AVG | **0.68** | 0.74 | 0.11 | 0.60 | **0.14** | **0.34** |
| Tiny ImageNet | SIG | 0.53 | 0.80 | 0.06 | 0.38 | 0.13 | 0.07 |
| | ISSBA | 0.53 | 0.95 | 0.02 | 0.44 | 0.25 | 0.18 |
| | Blended | 0.52 | 0.98 | 0.01 | 0.41 | 0.06 | 0.24 |
| | AVG | **0.53** | 0.91 | 0.03 | 0.41 | **0.15** | **0.16** |

### B. Analysis of Lightweight Retraining

The retraining process has significant lightweight features, which not only reduce computational overhead but also help to improve defense efficiency.

*1) From the Model Perspective:* The original suspicious model itself is already converging, and the retraining process is essentially a fine-tuning of the already converged model, rather than starting completely from scratch, and thus, the retraining process converges quickly.

*2) From the Sample Perspective:* Due to the small disturbance applied during the construction of retraining samples, the samples used in the retraining process are highly similar to the training samples of the original model. At the same time, the number of retrained samples is relatively small. Therefore, the training cost is relatively small.

*3) From the Optimization Principle:* According to deep learning theory, nodes near the input layer are mainly responsible for capturing the basic components of the image and simple low-level features, which are usually less correlated with specific backdoor triggers. Our goal is to deepen the model's ability to recognize high-level semantic features during the retraining process. Freezing these neurons related to low-level features can promote rapid adaptation of the retraining process to the new training set, making it more focused on our goals and reducing the consumption of computing resources.

### C. Limitations Analysis

The proposed defense scheme has certain limitations.

First, it requires a certain number of samples, which may still be a challenge in practice. Fortunately, we only need a small number of samples. If samples cannot be found, model inversion [46], [47] can also be used to obtain them.

Second, it focuses on backdoor attacks. As for whether the repaired model can resist other types of attacks, due to space limitations, this article will not elaborate on it.

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TABLE II
PERFORMANCE COMPARISON OF DIFFERENT DEFENSE SCHEMES (GROUP 2)

| Dataset | backdoor | No Defense | | FT | | | FP | | | MCR | | | NAD | | | (Ours, $\beta = 0.1$) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ACC | ASR | ACC | ASR | RA | ACC | ASR | RA | ACC | ASR | RA | ACC | ASR | RA | ACC | ASR | RA |
| CIFAR-10 | SIG | 0.98 | 0.84 | **0.89** | 0.13 | **0.64** | 0.88 | 0.21 | 0.50 | 0.86 | 0.07 | 0.54 | **0.89** | **0.03** | **0.64** | **0.89** | 0.06 | **0.64** |
| | ISSBA | 0.99 | 0.93 | **0.92** | 0.04 | 0.74 | 0.91 | 0.19 | 0.46 | 0.88 | 0.04 | 0.65 | **0.92** | **0.03** | **0.75** | **0.92** | 0.05 | 0.70 |
| | Blended | 0.99 | 0.93 | **0.92** | 0.99 | 0.00 | **0.92** | **0.21** | **0.52** | 0.86 | 0.61 | 0.31 | 0.91 | 0.57 | 0.34 | **0.92** | 0.94 | 0.05 |
| | AVG2 | 0.99 | 0.90 | 0.91 | 0.39 | 0.46 | 0.90 | **0.20** | 0.49 | 0.87 | 0.24 | 0.50 | 0.91 | 0.21 | **0.58** | **0.91** | 0.35 | 0.46 |
| CIFAR-100 | SIG | 0.69 | 0.78 | **0.69** | 0.72 | 0.10 | 0.60 | **0.29** | **0.14** | **0.69** | 0.77 | 0.09 | **0.69** | 0.58 | 0.13 | **0.69** | 0.66 | 0.11 |
| | ISSBA | 0.69 | 0.99 | **0.67** | 0.74 | 0.13 | 0.58 | 0.39 | 0.18 | **0.67** | 0.99 | 0.00 | **0.67** | **0.22** | **0.36** | **0.67** | 0.64 | 0.17 |
| | Blended | 0.69 | 0.99 | 0.67 | 0.99 | 0.01 | 0.59 | **0.41** | **0.16** | **0.69** | 0.99 | 0.00 | 0.68 | 0.97 | 0.02 | 0.67 | 0.93 | 0.04 |
| | AVG2 | 0.69 | 0.92 | **0.68** | 0.82 | 0.08 | 0.59 | **0.36** | 0.16 | **0.68** | 0.92 | 0.03 | **0.68** | 0.59 | **0.17** | **0.68** | 0.74 | 0.11 |
| Tiny ImageNet | SIG | 0.57 | 0.79 | **0.56** | 0.81 | **0.07** | 0.51 | **0.71** | **0.07** | 0.57 | 0.83 | 0.06 | 0.47 | 0.74 | 0.05 | 0.53 | 0.80 | 0.06 |
| | ISSBA | 0.56 | 0.99 | **0.55** | 0.86 | 0.07 | 0.49 | 0.92 | 0.03 | **0.55** | 0.99 | 0.00 | 0.47 | **0.35** | **0.15** | 0.53 | 0.95 | 0.02 |
| | Blended | 0.56 | 0.99 | 0.54 | 0.99 | 0.00 | 0.49 | **0.93** | **0.03** | **0.55** | 0.99 | 0.00 | 0.54 | 0.99 | 0.01 | 0.52 | 0.98 | 0.01 |
| | AVG2 | 0.56 | 0.92 | 0.55 | 0.89 | 0.05 | 0.50 | 0.85 | 0.04 | **0.56** | 0.94 | 0.02 | 0.49 | **0.69** | **0.07** | 0.53 | 0.91 | 0.03 |

TABLE III
IMPACT OF THE NOISE RATIO ON CIFAR-100 AND TINY-IMAGENET

| Dataset | backdoor | No Defense | | FT | | | FP | | | MCR | | | NAD | | | (Ours, $\beta = 0.1$ ) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ACC | ASR | ACC | ASR | RA | ACC | ASR | RA | ACC | ASR | RA | ACC | ASR | RA | ACC | ASR | RA |
| CIFAR-10 | BadNets | 0.95 | 0.91 | 0.88 | 0.02 | 0.87 | **0.91** | **0.01** | **0.90** | 0.84 | 0.02 | 0.84 | 0.89 | 0.02 | 0.88 | 0.90 | **0.01** | 0.88 |
| | TrojanNN | 0.99 | 0.93 | **0.92** | 0.12 | 0.75 | 0.91 | 0.46 | 0.38 | 0.87 | 0.14 | 0.66 | 0.91 | 0.08 | 0.78 | **0.92** | **0.03** | **0.85** |
| | WaNet | 0.97 | 0.90 | 0.92 | 0.16 | 0.76 | 0.92 | 0.04 | 0.87 | 0.92 | 0.03 | **0.88** | 0.92 | **0.02** | 0.87 | **0.93** | 0.04 | 0.87 |
| | AVG1 | 0.97 | 0.91 | 0.91 | 0.10 | 0.79 | 0.91 | 0.17 | 0.72 | 0.88 | 0.06 | 0.79 | 0.91 | 0.04 | 0.84 | **0.92** | **0.03** | **0.87** |
| CIFAR-100 | BadNets | 0.67 | 0.87 | 0.65 | **0.01** | 0.61 | 0.62 | **0.01** | 0.60 | **0.67** | 0.89 | 0.08 | 0.66 | **0.01** | 0.63 | 0.66 | **0.01** | **0.64** |
| | TrojanNN | 0.69 | 1.00 | 0.67 | 0.99 | 0.00 | 0.58 | 0.51 | 0.17 | **0.68** | 1.00 | 0.00 | 0.67 | 0.99 | 0.01 | **0.68** | **0.01** | **0.57** |
| | WaNet | 0.68 | 0.89 | 0.67 | 0.13 | 0.55 | 0.65 | 0.44 | 0.35 | 0.63 | 0.99 | 0.00 | 0.66 | **0.07** | **0.60** | **0.68** | 0.18 | 0.56 |
| | AVG1 | 0.68 | 0.92 | 0.66 | 0.38 | 0.39 | 0.62 | 0.32 | 0.37 | 0.66 | 0.96 | 0.03 | 0.66 | 0.36 | 0.41 | **0.67** | **0.07** | **0.59** |
| Tiny ImageNet | BadNets | 0.56 | 0.99 | 0.55 | 0.01 | 0.54 | 0.50 | 0.05 | 0.48 | **0.56** | 0.98 | 0.01 | 0.54 | **0.00** | **0.54** | 0.55 | **0.00** | **0.55** |
| | TrojanNN | 0.56 | 0.99 | 0.54 | 0.01 | 0.44 | 0.50 | 0.99 | 0.00 | 0.55 | 0.99 | 0.01 | 0.46 | 0.01 | 0.35 | **0.56** | **0.00** | **0.49** |
| | WaNet | 0.57 | 0.99 | 0.55 | **0.01** | **0.53** | 0.50 | 0.01 | 0.45 | **0.57** | 0.05 | 0.43 | 0.56 | 0.01 | **0.53** | **0.57** | **0.01** | 0.52 |
| | AVG1 | 0.56 | 0.99 | 0.55 | 0.01 | 0.50 | 0.50 | 0.35 | 0.31 | **0.56** | 0.67 | 0.15 | 0.52 | 0.01 | 0.47 | **0.56** | **0.00** | **0.52** |

In addition, it focuses on image-related models. Further research is needed to determine whether it is suitable for defending against backdoor attacks in other types of deep learning models, such as sequence and graph models.

## V. EXPERIMENTAL EVALUATION

### A. Experimental Setting

*1) Datasets and Evaluation Metrics:* We use ResNet-18 as the base model to train backdoor attack models on CIFAR-10, CIFAR-100, and Tiny ImageNet for subsequent evaluation. Evaluation metrics include ACC, RA, and ASR, whose full names can be found in Section III-C1.

*2) Backdoor Attack Schemes and Settings:* Six representative backdoor attack schemes were selected, including BadNets [2], TrojanNN [7], Blended [3], SIG [5], ISSBA [4], and WaNet [6]. According to the defense difficulty, we divide these six types of attack models into two groups: BadNets, TrojanNN, and WaNet as group 1, and ISSBA, SIG, and Blended as group 2. In the training process of all backdoor attack models, we set the label of the backdoor trigger sample to 0, the backdoor trigger sample injection rate to 0.1, and the learning rate to 0.01. On the CIFAR-10 dataset containing backdoor triggers, the training epoch is set to 100. On the CIFAR-100 and ImageNet subsets containing backdoor triggers, the train epoch is set to 200.

*3) Backdoor Defense Schemes and Settings:* Four representative backdoor defense schemes were selected, including FT [23], FP [23], mode connectivity repair (MCR) [25], and neural attention distillation (NAD) [26]. We will follow the original configuration and recommendations of these schemes. For FP, we pruned the last convolutional layer of the model. For MCR, we trained a loss curve of 100 epochs using a backdoor model as the endpoint and evaluated the model's defense performance on the loss curve. For NAD, we trained 10 epochs in the student model under the guidance of the teacher model. We have set the same retraining round for all defense schemes. When these schemes require clean samples for training or fine-tuning, we use 2% of the training dataset. In addition, data augmentation (including random cropping and random flipping) is also applied in all defense schemes.

### B. Comparison of Defense Performance

This evaluation sets the noise ratio $\beta$ to 0.1, which is a mild defense configuration that can defend against general backdoor attacks. Tables I and II show the comparison results. Each table involves three datasets (column 1): CIFAR-10, CIFAR-100, and Tiny ImageNet. Six different types of backdoor attack models (column 2) are constructed based on each dataset. The 3rd column (no defense) contains two metrics (ACC and ASR) as no defense scheme is applied. Columns 4–8 contain three metrics: ACC, ASR, and RA. Among them,

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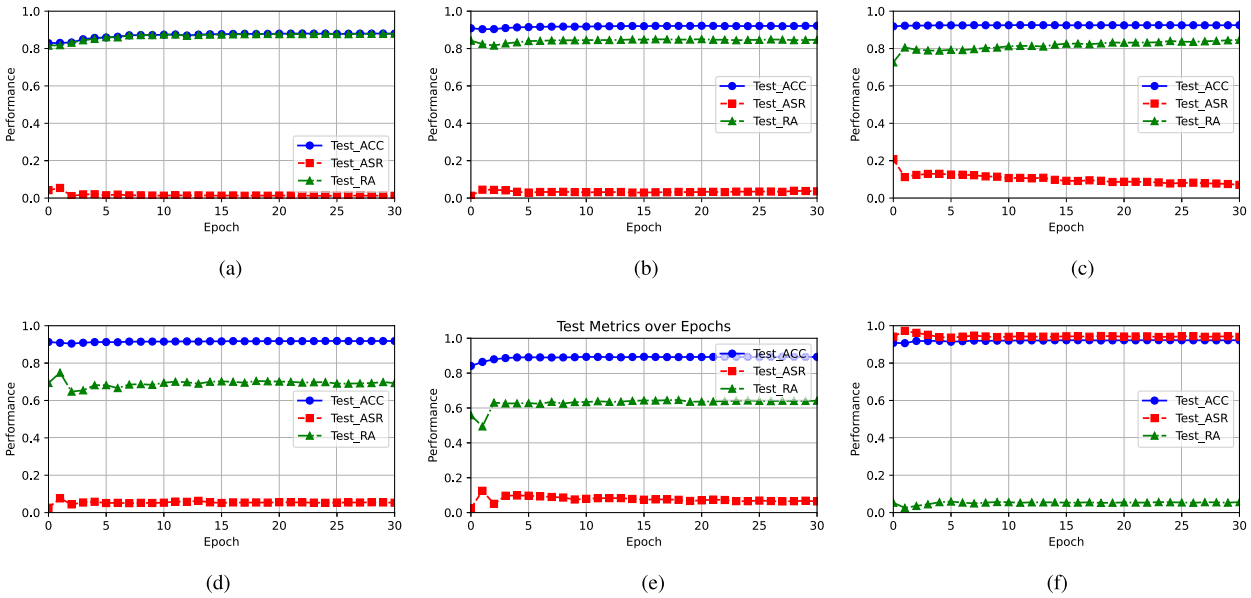



Fig. 7. Impact of retraining rounds on defense performance. (a) BadNets. (b) TrojanNN. (c) WaNet. (d) ISSBA. (e) SIG. (f) Blended.

column 8 (ours) shows the performance of the proposed scheme. *AVG*1 and *AVG*2 are the average performance results on two groups.

According to Table I, the performance of each defense schemes are well enough in the backdoor attack models of group 1 (i.e., BadNets, TrojanNN, and WaNet), where the proposed scheme achieves better performance than other schemes. Taking the CIFAR-10 dataset as an example, according to the *AVG*1 (row 6), the proposed scheme achieves the best performance. With ASR decreasing from 0.91 (column 4) to 0.03 (column 18) and RA reaching 0.87 (column 19), it indicates that the defense performance of the proposed scheme is significant. At the same time, ACC is 0.92 (column 17), which is slightly lower than 0.97 (column 3). This indicates that the implementation of the proposed scheme usually has a small negative impact on the prediction accuracy of clean samples, but the accuracy is still very high.

According to Table II, under a mild defense configuration ($\beta = 0.1$), the proposed scheme does not achieve the best performance in these more covert backdoor attack models (i.e., SIG, ISSBA, and Blended). At the same time, the defense performance of all defense schemes in group 2 is not satisfactory. Taking ASR as an example, FP achieved the best ASR value (column 9) on CIFAR-10 and CIFAR-100, with 0.20 and 0.36, respectively. On the Tiny ImageNet, NAD achieved the best ASR value (0.69, column 15).

### C. Impact of the Noise Ratio on Defense Performance

This section investigates the impact of the noise ratio $\beta$. According to Section V-B, when using the mild defense configuration ($\beta = 0.1$), the defense performance in group 1 is already good enough. Therefore, we will focus on Group 2 (i.e., SIG, ISSBA, and Blended). At the same time, the performance of the proposed scheme in CIFAR-10 is quite different from that in CIFAR-100 and Tiny-ImageNet, so we will investigate them separately.

*1) CIFAR-10:* On the CIFAR-10 dataset, the proposed scheme has a good defense performance against ISSBA and SIG. Therefore, we only conducted further analysis on Blended and attempted to explore the impact of the noise ratio on defense performance. According to Fig. 5, as the noise ratio is adjusted from a mild defense configuration ($\beta = 0.1$) to an enhanced defense configuration ($\beta = 1$), the defense performance on stealthier attacks is significantly improved. The decrease in ACC is extremely slight, while the decrease in ASR and the increase in RA are significantly faster. For example, when the noise ratio increases to 0.8, ASR has already decreased to 0.2, while RA has increased to 0.63, and the change in ACC is very small.

*2) CIFAR-100 and Tiny-ImageNet:* According to Section V-B, the proposed scheme has unsatisfactory defense performance against ISSBA, SIG, and Blended attack models constructed on the CIFAR-100 and Tiny ImageNet. We will re-evaluate the impact of the noise ratio parameter on defense performance based on these three attack models.

According to Table III, from the perspective of average value (row 6 and 10), when the noise ratio parameter increases to 1, the ASR on the CIFAR-100 dataset decreases significantly from the original 0.74 (column 4) to 0.14 (column 7). This means that the success attack probability of the adversary is significantly reduced, and the defense performance is greatly improved. At the same time, the RA value has also increased from 0.11 (column 5) to 0.34 (column 8). Comparing the RA values of other defense schemes in Table II, it is a very significant improvement. Of course, the performance improvement also comes at a certain cost, and the ACC value has also decreased from 0.68 (column 3) to 0.60 (column 6). Table III also reveals that the performance improvement degree varies among different attack models, with Blended showing

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TABLE IV
COMPARISON OF COMPUTATIONAL COST

| Schemes | FT | FP | NAD | MCR | OURS |
|---|---|---|---|---|---|
| Time (minute) | 13.07 | 23.56 | 4.53 | 54.82 | 3.25 |

the most significant improvement effect. The performance improvement on the Tiny ImageNet dataset is similar to that on the CIFAR-100 dataset.

### D. Impact of the Sample Ratio on Defense Performance

This section takes the BadNets model on the CIFAR-10 as an example to evaluate the impact of the sample ratio ($\rho$). According to Fig. 6, as the proportion of the original dataset increases, the ASR gradually decreases, while the ACC and RA increase to a certain extent. The improvement of defense performance does not show linear growth. When the sample ratio is large (e.g., greater than 0.06), the improvement of ACC, RA, and ASR slows down, and the marginal benefits significantly decrease.

### E. Impact of Retraining Rounds on Defense Performance

This experiment is based on CIFAR-10. We conducted 100 epochs of retraining on the backdoor attack models by using the proposed scheme, and recorded the ACC, ASR, and RA after each epoch. Only the first 30 epochs are retained on the horizontal axis in Fig. 7, due to the minimal changes. Among all backdoor attack models, in most cases, our defense scheme only needs to be retrained for two to three epochs, and its performance curve converges to a relatively stable value.

### F. Comparison of Computational Cost

We record the implementation time required for different defense methods against the BadNet model constructed based on CIFAR-10. The time units of Table IV are in minutes. The basic information of the experimental platform is as follows: Intel Xeon[1] Gold 6226R at 2.9 GHz * 32 processor, 64 GB memory, GeForce GTX 3090 graphics card. Compared to other defense schemes, the proposed scheme has a significant competitive advantage in computational cost.

## VI. CONCLUSION

This article introduces frequency-domain perturbations into backdoor defense scenarios and proposes a lightweight backdoor defense scheme to demonstrate it. In the repaired model, the coupling relationship between the gate triggering mode and the target category is weakened. Theoretical analysis and experimental evaluation indicate that our defense strategy applies to various backdoor attack scenarios and has comparative advantages in multiple evaluation metrics, with high universality and applicability.

[1]Registered trademark.

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